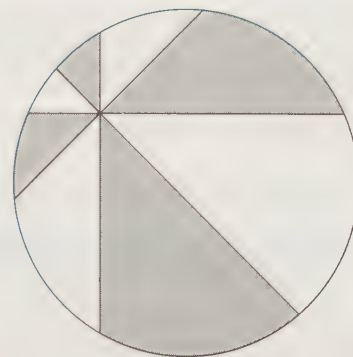




The Pizza Theorem

There is no transfer into another kind, like the transfer from length to area and from area to a solid. – Aristotle

CHAPTER 4

Perimeter and Area

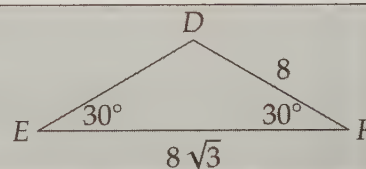
4.1 Perimeter

The **perimeter** of a closed figure is how far you travel if you walk along its boundary all the way around it once. So, the perimeter of a triangle is simply the sum of the lengths of the sides of the triangle.

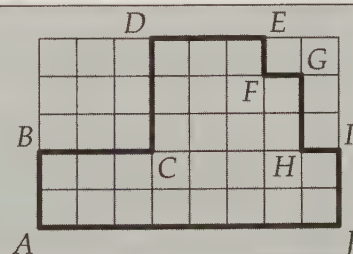
Problems

Problem 4.1: $\triangle ABC$ is equilateral. Find AB given that the perimeter of $\triangle ABC$ is 36.

Problem 4.2: Find the perimeter of $\triangle DEF$.



Problem 4.3: Given that each little square in the grid is a 1×1 square, find the perimeter of $ABCDEFGHIJ$.



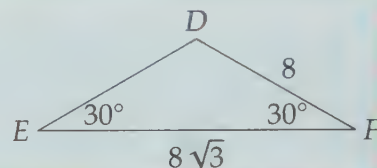
Problem 4.4: The length of each leg of an isosceles triangle is three times the length of the base of the triangle. The perimeter of the triangle is 91. What is the length of the base of the triangle?

Questions about perimeter are usually just questions about segment lengths. We'll try a few.

Problem 4.1: $\triangle ABC$ is equilateral. Find AB given that the perimeter of $\triangle ABC$ is 36.

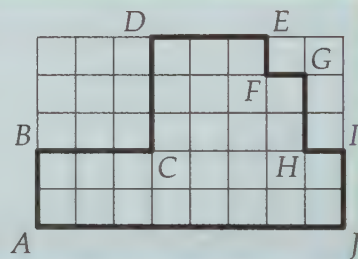
Solution for Problem 4.1: Since $\triangle ABC$ is equilateral, all of its sides are equal. Let each have length s . Then the perimeter is $3s$. Therefore, we have $3s = 36$, so $s = 12$. $\square$

Problem 4.2: Find the perimeter of $\triangle DEF$.



Solution for Problem 4.2: Since $\angle E = \angle F$, we have $DE = DF = 8$. Therefore, our perimeter is $DE + DF + EF = 16 + 8\sqrt{3}$. $\square$

Problem 4.3: Given that each little square in the grid is 1×1 square, find the perimeter of $ABCDEFGHIJ$.



Solution for Problem 4.3: We could simply find the lengths of the 10 sides and add up all the numbers, but there's a slicker approach. All of our sides are either vertical or horizontal. The horizontal side on the bottom has length 8; therefore, the sum of the horizontal sides on top, which together cover the same horizontal distance without backtracking, must be 8. Similarly, $AB + CD = 5$, so the vertical sides on the left have total length 5. The ones on the right have the same total length, so they contribute 5 to the perimeter as well. So, our perimeter is $2(8 + 5) = 26$. $\square$

Problem 4.4: The length of each leg of an isosceles triangle is three times the length of the base of the triangle. The perimeter of the triangle is 91. What is the length of the base of the triangle?

Solution for Problem 4.4: Let the base of the triangle have length x . Then the length of each leg is $3x$. Since the perimeter of the triangle is 91, we must have $x + 3x + 3x = 91$. Solving this equation for x , we find $x = 13$. The length of the base of the triangle is 13. $\square$

Concept: The key to solving most geometric word problems is the same as for most non-geometric word problems: assign variables to unknown quantities and use the words to make equations you can solve.

Exercises

4.1.1 In the diagram at left below, $XY = 5\sqrt{2}$, $YZ = 10$, and $\angle Y = \angle Z = 45^\circ$. Find the perimeter of $\triangle XYZ$.

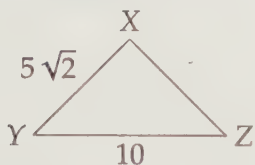


Figure 4.1: Diagram for Problem 4.1.1

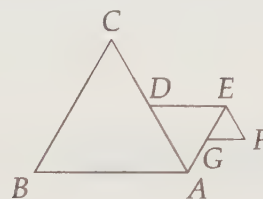


Figure 4.2: Diagram for Problem 4.1.2

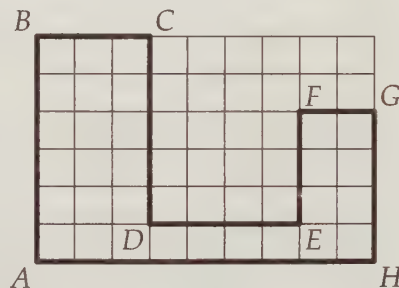
4.1.2 Triangles ABC , ADE , and EFG at right above are all equilateral. Points D and G are midpoints of $\overline{AC}$ and $\overline{AE}$, respectively. If $AB = 4$, what is the perimeter of figure $ABCDEFG$? (Source: AMC 8)

4.1.3

- Must two congruent triangles have the same perimeter? Why or why not?
- Must two triangles with the same perimeter be congruent? Why or why not?
- ★ Must two triangles with the same perimeter be congruent if the triangles have one angle measure in common?

4.1.4 Each little square in the grid at right is a 1×1 square. Find the perimeter of $ABCDEFGH$.

4.1.5 A triangle with perimeter 45 has one side that is twice as long as the shortest side and another side that is 50% longer than the shortest side. Find the length of the shortest side of the triangle. **Hints:** 480



4.2 Area

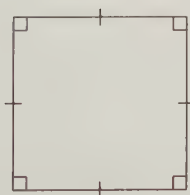


Figure 4.3: A Square

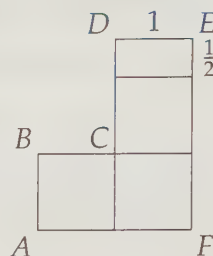


Figure 4.4: A Rectangle

A **square** is a figure that has four sides equal in length and four right angles, as shown in Figure 4.3. We call a four-sided figure with four right angles a **rectangle**, an example of which is shown in Figure 4.4. (A square is a special type of rectangle.) The opposite sides of a rectangle have equal length, as shown.

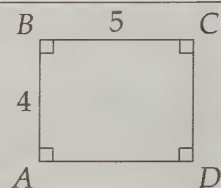
We'll learn more about squares and rectangles in Chapter 8.

Roughly speaking, the **area** of a figure is the total number of 1×1 squares needed to cover the figure, where we add fractional squares when we only need a piece of a square to do the job. For example, we need $3\frac{1}{2}$ squares to cover the region $ABCDEF$ at right, so we say the area of $ABCDEF$ is $3\frac{1}{2}$. We will often use brackets to denote area, like this: $[ABCDEF] = 3\frac{1}{2}$.



Problems

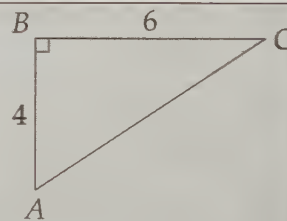
Problem 4.5: Consider rectangle $ABCD$, which has $AB = CD = 4$, $BC = AD = 5$, and right angles at A , B , C , and D . What is the area of $ABCD$?



Problem 4.6: The length of one side of a rectangle is 4 less than 3 times an adjacent side. The perimeter of the rectangle is 64. Find the area of the rectangle.

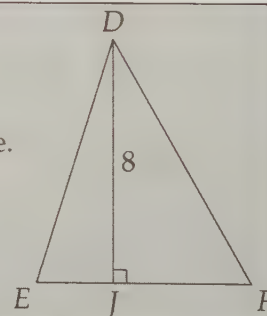
Problem 4.7: A triangle that has a right angle as one of its angles is called a **right triangle**.

- Find the area of right triangle $\triangle ABC$, where $AB = 4$, $BC = 6$, and $\angle B = 90^\circ$.
- Find a formula for the area of a right triangle given the lengths of its sides.

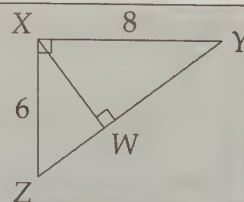


Problem 4.8:

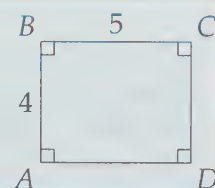
- Find the area of $\triangle DEF$ given that J is on $\overline{EF}$, $DJ = 8$, $\overline{DJ} \perp \overline{EF}$, and $EF = 7$.
- Use the previous part to come up with a way to find the area of any triangle. Does your approach work for obtuse triangles?



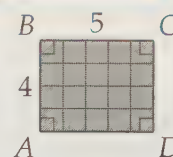
Problem 4.9: Given that $\angle ZXY = \angle XWY = 90^\circ$, $XY = 8$, $XZ = 6$, and $ZY = 10$, find XW .



Problem 4.5: Consider rectangle $ABCD$, which has $AB = CD = 4$, $BC = AD = 5$, and right angles at A , B , C , and D . What is the area of $ABCD$?



Solution for Problem 4.5: We can fill $ABCD$ with four rows of five 1×1 squares each, as shown to the right. So, the area is $(4)(5) = 20$. $\square$



Finding the area of a rectangle by taking the product of consecutive sides works even when the sides aren't integers.

Important: A rectangle with two adjacent sides of lengths ℓ and w has area ℓw . Since a square is just a rectangle in which these lengths are the same, the area of a square equals the square of the length of one of its sides.



Let's apply our knowledge of rectangles to a problem.

Problem 4.6: The length of one side of a rectangle is 4 less than 3 times an adjacent side. The perimeter of the rectangle is 64. Find the area of the rectangle.

Solution for Problem 4.6: Let the lengths of the two sides be x and y . We are given that $x = 3y - 4$. Since the opposite sides of a rectangle are equal, the perimeter of the rectangle is $2x + 2y = 2(3y - 4) + 2y = 8y - 8$. We are given that the perimeter is 64, so we have $8y - 8 = 64$. Therefore, $y = 9$ and $x = 3(9) - 4 = 23$, so the area of the rectangle is $9(23) = 207$. $\square$

Now let's use our rectangle formula to find the area of a triangle. We'll start with a special kind of triangle.

Problem 4.7: A triangle that has a right angle as one of its angles is called a **right triangle**.

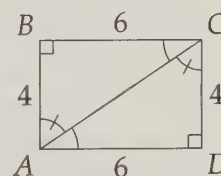
- Find the area of right triangle $\triangle ABC$, where $AB = 4$, $BC = 6$, and $\angle B = 90^\circ$.
- Find a formula for the area of a right triangle given the lengths of its sides.

Solution for Problem 4.7:

- We don't know anything about finding the area of a triangle yet; all we know about are rectangles. However, we know that if we draw a diagonal of a rectangle, we cut the rectangle into two congruent right triangles. In this problem we have a triangle – perhaps we can make a rectangle by just adding another right triangle as shown below.

In the diagram, $\triangle ACD$ is a copy of $\triangle CAB$. Since $\angle BAC + \angle B + \angle BCA = 180^\circ$ and $\angle B = 90^\circ$, we have $\angle BAC + \angle BCA = 90^\circ$. Since $\triangle CAD \cong \triangle ACB$, we have $\angle DAC = \angle BCA$. Therefore, $\angle BAC + \angle DAC = 90^\circ$, so $\angle BAD$ is a right angle. Similarly, $\angle BCD$ is a right angle, so $ABCD$ is a rectangle.

Since $ABCD$ is a rectangle with sides 4 and 6, we know that $[ABCD] = 24$. Our two right triangles are congruent and the sum of their areas is $[ABCD]$. So, each right triangle has area $24/2 = 12$.



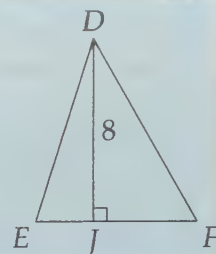
- (b) Let a and b be the lengths of the two sides adjacent to the right angle of $\triangle ABC$. We call the sides adjacent to a right angle in a right triangle the **legs** of the right triangle. As shown in the previous part, we can build a rectangle with consecutive sides of lengths a and b out of two congruent copies of $\triangle ABC$. The rectangle then has area ab , so each of the two triangles has area $ab/2$. So, the area of any right triangle is half the product of the lengths of the legs of the triangle.

□

So that takes care of right triangles. But what if the triangle isn't a right triangle?

Problem 4.8:

- (a) Find the area of $\triangle DEF$ given that J is on $\overline{EF}$, $DJ = 8$, $\overline{DJ} \perp \overline{EF}$, and $EF = 7$.
 (b) Find a method to determine the area of any triangle.



Solution for Problem 4.8:

- (a) We do know how to tackle right triangles, so we break $\triangle DEF$ into $\triangle DEJ$ and $\triangle DJF$:

$$[DEF] = [DEJ] + [DJF] = \frac{(EJ)(DJ)}{2} + \frac{(JF)(DJ)}{2} = \frac{DJ}{2}(EJ + JF) = \frac{DJ}{2}(EF) = \left(\frac{8}{2}\right)(7) = 28.$$

- (b) We call a perpendicular segment from a vertex of a triangle to the opposite side of the triangle an **altitude** of the triangle. Sometimes an altitude is also called a **height**. The terms 'altitude' and 'height' can also refer to the length of the altitude. The side to which the altitude is drawn is called the **base** that corresponds to the given altitude. For example, we say $\overline{DJ}$ is the altitude to base $\overline{EF}$ in $\triangle DEF$ above. (As with 'altitude', the term 'base' can also refer to the length of the base.) Point J gets a special name, too. The point at which an altitude meets the base to which it is drawn is sometimes called the **foot** of the altitude. Thus, J is the foot of the altitude from D to $\overline{EF}$.

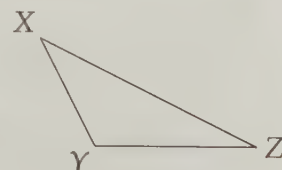
Our work so far strongly suggests that the area of a triangle is one-half the product of an altitude of a triangle and the length of the base to which the altitude is drawn. For example, in part (a), the area is one-half the product of EF and DJ , and $\overline{DJ}$ is the altitude from D to base $\overline{EF}$.

We have already shown that if our base is either leg of a right triangle, then the area is

$$\frac{\text{base} \times \text{altitude}}{2}.$$

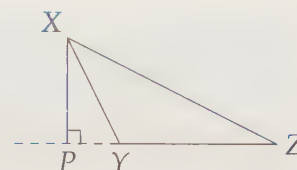
Our work on the first part of this problem can be used to show that this formula works whenever the two angles at the base are acute, as $\angle E$ and $\angle F$ are.

But what if one of the angles at the base is obtuse? In the diagram to the right, let $\overline{YZ}$ be the base. Here, we can't draw a line from X that is perpendicular to $\overline{YZ}$, so we extend side $\overline{YZ}$ and draw our altitude to the extension as shown on the next page. Now we need to check whether $[XYZ] = (XP)(YZ)/2$.



In our first part we added areas; here, we subtract:

$$\begin{aligned}
 [XYZ] &= [XPZ] - [XPY] \\
 &= \frac{(XP)(PZ)}{2} - \frac{(XP)(PY)}{2} \\
 &= \frac{XP(PZ - PY)}{2} \\
 &= \frac{(XP)(YZ)}{2}.
 \end{aligned}$$



We have to tweak our definition of 'altitude' a little, but our formula still works.

□

We can now summarize our findings about the area of a triangle.

Important:

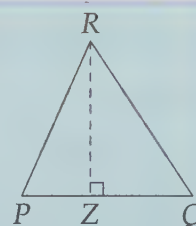


To find the area of a triangle, we select one side to be the base. The perpendicular segment from the vertex opposite the base to the base (extended if necessary) is the **altitude**. We then have:

$$\text{Area} = \frac{\text{base} \times \text{altitude}}{2}.$$

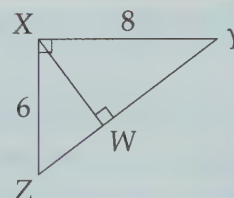
For example, in the triangle shown, we have:

$$[PQR] = \frac{PQ \times RZ}{2}.$$



You'll get plenty of practice finding areas of triangles in the pages to come. But first, we see that unlike perimeter, which is rarely a useful problem solving tool, area can be used to solve problems that don't appear to have anything to do with area.

Problem 4.9: Given that $\angle ZXY = \angle XWY = 90^\circ$, $XY = 8$, $XZ = 6$, and $ZY = 10$, find XW .



Solution for Problem 4.9: This problem only has information about lengths and angles, and it doesn't even ask for area. However, all those perpendicular lines make us think about area. We can quickly see that

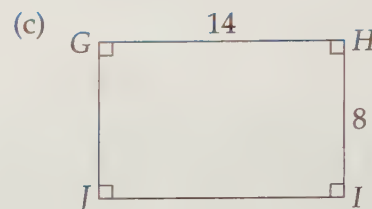
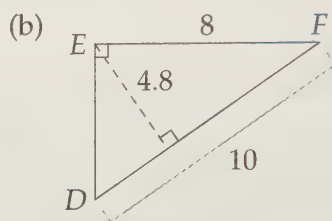
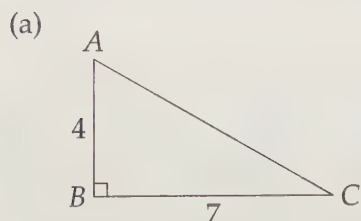
$$[XYZ] = \frac{(XY)(XZ)}{2} = 24.$$

$\overline{XW}$ is the altitude to side $\overline{YZ}$, so $[XYZ] = (XW)(YZ)/2$. Since $[XYZ] = 24$, we have the equation $24 = (XW)(10)/2 = 5XW$, so $XW = 24/5$. □

Sprinkled throughout this book will be many other uses of area as a problem solving tool.

Exercises

4.2.1 Find the area of each figure shown below.

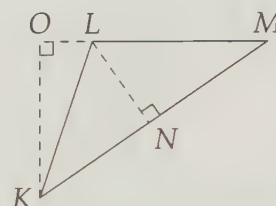


4.2.2 What is the area of square $ABCD$ if the perimeter of the square is 36?

4.2.3 In the diagram at right, $KO = 6$, $LM = 7$, and $KM = 12$.

(a) Find $[KLM]$.

(b) Find LN . **Hints:** 506



4.2.4 One gallon of paint covers 80 square feet of wall. How many gallons of paint do I have to buy if I am going to paint two rectangular walls that are each 24 feet long and 8 feet high?

4.2.5 A square poster is replaced by a rectangular poster that is 2 inches wider and 2 inches shorter. What is the difference in the number of square inches between the area of the larger poster and the smaller poster? (Source: MATHCOUNTS) **Hints:** 102

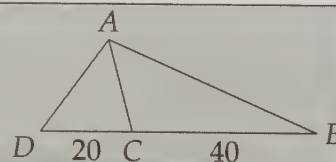
4.2.6★ The perimeter of a square garden is 64 meters. The path surrounding the garden has uniform width and has an area of 228 square meters. How many meters of fencing are needed to surround the outer edge of the path? (Source: MATHCOUNTS) **Hints:** 494



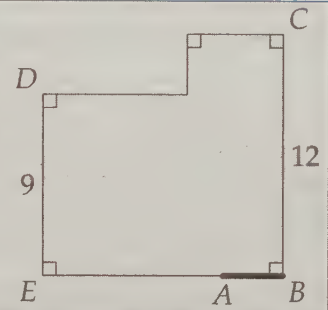
4.3 Same Base/Same Altitude

Problems

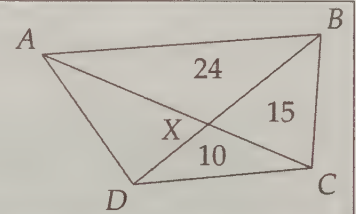
Problem 4.10: Given that $BC = 40$ and $CD = 20$ in the figure, what are $[ABC]/[ACD]$ and $[ABC]/[ABD]$?



Problem 4.11: Suba and Sam have been hired to paint a triangle with base $\overline{AB}$ on the weirdly shaped wall in the diagram. They are told to choose either C or D as the third vertex of the triangle. Suba thinks using D will make the triangle look cooler, but Sam thinks using C will result in a smaller triangle to paint. How can Suba convince Sam to choose D ?

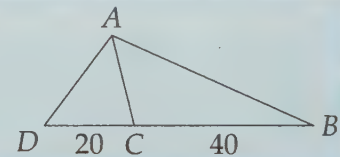


Problem 4.12: $\overline{AC}$ and $\overline{BD}$ meet at X as shown. Given $[ABX] = 24$, $[BCX] = 15$, and $[CDX] = 10$, find $[ADX]$.



In this section we examine a particularly useful triangle area concept.

Problem 4.10: Given that $BC = 40$ and $CD = 20$ in the figure, what are $[ABC]/[ACD]$ and $[ABC]/[ABD]$?



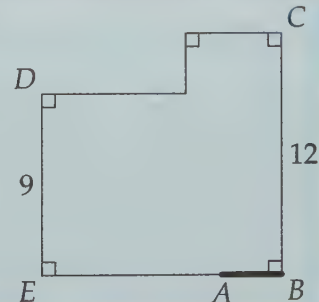
Solution for Problem 4.10: The altitude from A is the same for both $\triangle ACD$ and $\triangle ABC$. Let h be the length of this altitude. Hence, $[ABC] = (BC)(h)/2 = 20h$ and $[ACD] = (CD)(h)/2 = 10h$. Therefore, $[ABC]/[ACD] = 20h/10h = 2$. Notice that $[ABC]/[ACD] = (BC)/(CD)$.

Similarly, the altitude from A in each of $\triangle ABC$ and $\triangle ABD$ has length h . Therefore,

$$\frac{[ABC]}{[ABD]} = \frac{(BC)(h)/2}{(BD)(h)/2} = \frac{BC}{BD} = \frac{2}{3}.$$

A key step in our solution is noting that the three triangles have an altitude in common. Notice that in each case the ratio of the areas of the triangles equals the ratio of the lengths of the sides to which this common altitude is drawn. $\square$

Problem 4.11: Suba and Sam have been hired to paint a triangle with base $\overline{AB}$ on the weirdly shaped wall in the diagram. They are told to choose either C or D as the third vertex of the triangle. Suba thinks using D will make the triangle look cooler, but Sam thinks using C will result in a smaller triangle to paint. How can Suba convince Sam to choose D ?



Solution for Problem 4.11: To figure out which of $\triangle ABC$ or $\triangle ABD$ will use more paint, we must consider the areas of the two triangles. Since

$$[ABD] = \frac{(AB)(DE)}{2} \quad \text{and} \quad [ABC] = \frac{(AB)(BC)}{2},$$

we see that

$$\frac{[ABD]}{[ABC]} = \frac{(AB)(DE)/2}{(AB)(BC)/2} = \frac{DE}{BC} = \frac{3}{4}.$$

So, Suba should point out that $\triangle ABD$ will require less paint because its area is smaller than that of $\triangle ABC$. $\square$

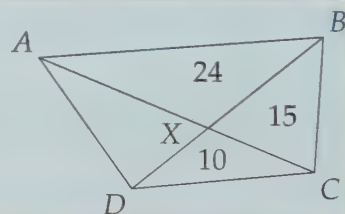
The last two problems illustrate two useful problem solving principles:

Important:



1. If two triangles share an altitude, then the ratio of their areas is the ratio of the bases to which that altitude is drawn. This is particularly useful in problems in which two triangles have bases along the same line.
2. If two triangles share a base, then the ratio of their areas is the ratio of the altitudes to that base.

Problem 4.12: $\overline{AC}$ and $\overline{BD}$ meet at X as shown. Given $[ABX] = 24$, $[BCX] = 15$, and $[CDX] = 10$, find $[ADX]$.



Solution for Problem 4.12: Since $\triangle ABX$ and $\triangle CBX$ share an altitude from B , we have

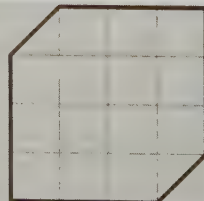
$$\frac{AX}{CX} = \frac{[ABX]}{[CBX]} = \frac{8}{5}.$$

Turning to triangles $\triangle ADX$ and $\triangle CDX$, we have

$$\frac{[ADX]}{[CDX]} = \frac{AX}{CX} = \frac{8}{5}.$$

Therefore, $[ADX] = (8/5)[CDX] = 16$. $\square$

Extra! Can you cut the figure at right into two congruent pieces that can be arranged to form a rectangle?



Exercises

4.3.1 In the diagram at left below, $PT = 6$, $TR = 3$, and $QV = 4$.

- Find $[PQR]$, $[PTQ]$, and $[QTR]$.
- Find TR/PT and $[QTR]/[PTQ]$.

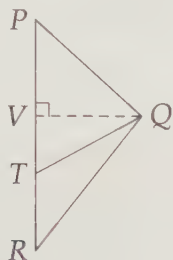


Figure 4.5: Diagram for Problem 4.3.1

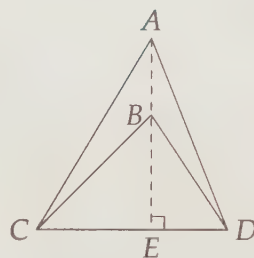


Figure 4.6: Diagram for Problem 4.3.2

4.3.2 In the diagram at right above, $AB = 2$ and $BE = 3$.

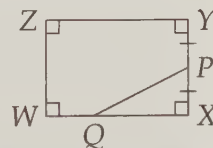
- Find $[BCD]/[ACD]$.
- Find $[ADBC]/[ADC]$. **Hints:** 255
- Find $[BCD]/[ACBD]$. **Hints:** 432

4.3.3 Euclid Apartment Building has a wall that is 30 feet long and 10 feet high. Jean is supposed to paint a triangle with one vertex at the top of the wall, and a base that runs 8 feet along the bottom of the wall. Jean wants to put the vertex at one corner and the base at the other side of the wall as shown on the left. George, the building owner, wants to save money on paint. He insists that the top vertex be right above the middle of the base, as shown on the right, so the triangle won't be so big. However, to show that he's a nice guy, he says that she can make the base 10 feet wide instead. Jean argues that George's design will use more paint than hers. Is she right?



4.3.4 By what factor is the area of triangle multiplied if the length of its base is doubled and the height is tripled? (Source: MATHCOUNTS)

4.3.5★ Find QX in the diagram at right given that $WX = 8$ and $[PQX] = [WXYZ]/6$.
Hints: 244, 390



Extra! For every complex problem there is an answer that is clear, simple, and wrong.



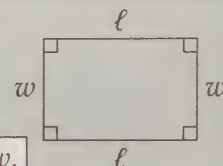
—H. L. Mencken

4.4 Summary

Definition: The **perimeter** of a closed figure is how far you travel if you walk along its boundary all the way around it once.

Definition: The **area** of a closed figure is the number of 1×1 squares (or pieces of squares) needed to exactly cover the figure. We sometimes use brackets to denote area, so that $[ABC]$ means the area of $\triangle ABC$.

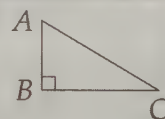
Definition: A **rectangle** has four sides and four right angles, as shown. Furthermore, opposite sides of a rectangle equal each other in length.



Important: The area of a rectangle with consecutive sides of length ℓ and w is ℓw . Since a **square** is just a rectangle in which these lengths are the same, the area of a square equals the square of the length of one of its sides.



Definitions: A triangle that has a right angle as one of its angles is called a **right triangle**. The sides adjacent to the right angle of a triangle are called the **legs** of the triangle.

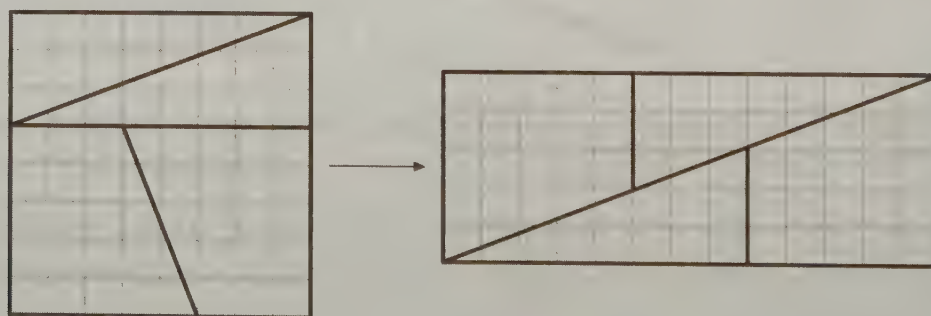


Important: The area of a right triangle is half the product of the legs of the triangle. For example, in the right triangle shown, we have

$$[ABC] = \frac{(AB)(BC)}{2}.$$



Extra! Tedrick, trying to trick me into believing that 8×8 is 65, showed me how he could cut an 8×8 square into 4 pieces that he could then reassemble into a 5×13 rectangle.



The 8×8 square has an area of 64, but the 5×13 rectangle has an area of 65. How is this possible?

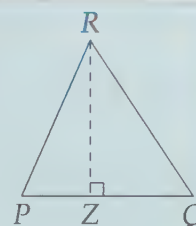
Important:


To find the area of a triangle, we select one side to be the base. The perpendicular segment from the vertex opposite the base to the base (extended if necessary) is the **altitude**. We then have:

$$\text{Area} = \frac{\text{base} \times \text{altitude}}{2}.$$

For example, in the triangle shown, we have:

$$[PQR] = \frac{PQ \times RZ}{2}.$$



Area can be a very powerful problem-solving tool. One particularly useful pair of principles is:

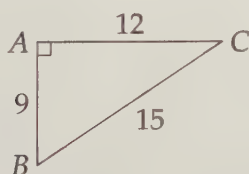
Important:


1. If two triangles share an altitude, the ratio of their areas is the ratio of the bases to which that altitude is drawn. This is particularly useful in problems in which two triangles have bases along the same line.
2. If two triangles share a base, then the ratio of their areas is the ratio of the altitudes to that base.

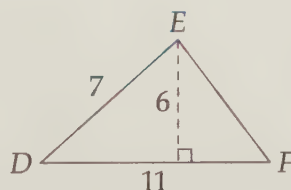
REVIEW PROBLEMS

4.13 Find the area of each figure below.

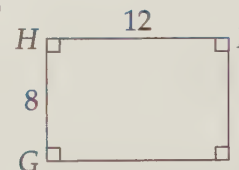
(a)



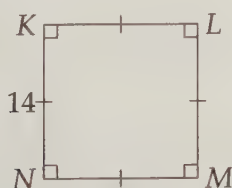
(b)



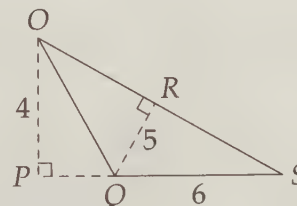
(c)



(d)



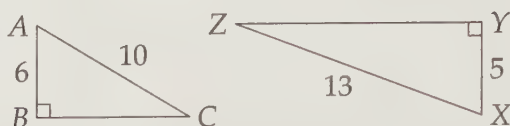
(e)



4.14 A triangle has area 42 and one side of length 7. Find the length of the altitude to the side of length 7. Is it possible to find the perimeter of this triangle?

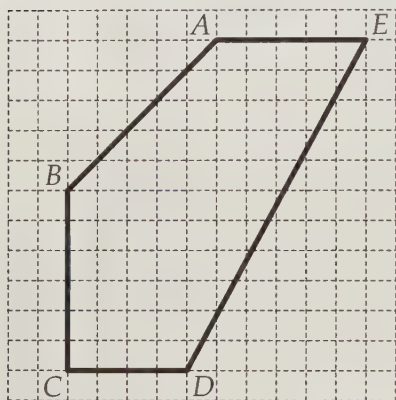
4.15 Find the perimeter of a square that has an area of 75.

4.16 Find the perimeters of $\triangle ABC$ and $\triangle XYZ$ shown below given that $[ABC] = 24$ and $[XYZ] = 30$.



4.17 The length of a given rectangle is 2 less than 4 times the width of the rectangle. Its perimeter is 51. What is its area?

4.18 I have a rectangular living room that is 24 feet long and 16 feet wide. I want to buy a carpet that costs \$2.50 per square foot, and I want to leave a 2 foot margin between the carpet and the wall all the way around the room. How much will the carpet cost?



4.19 What is the number of square units enclosed by the figure $ABCDE$?

4.20 A rectangle has a perimeter of 28 cm and an area of 48 cm^2 . Find the dimensions of the rectangle.

4.21 If the height of a triangle is multiplied by 4, what must we do to the base of the triangle in order to leave the area unchanged?

4.22 The altitude from A to $\overline{BC}$ in $\triangle ABC$ has the same length as the altitude from B to $\overline{AC}$. Prove that $BC = AC$.

4.23 The area of rectangle $ABCD$ shown at left below is 36 and $DE = 2EC$.

(a) What is $[BCD]$?

(b) What is $[BED]$?

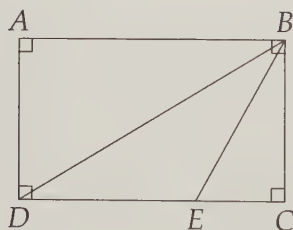


Figure 4.7: Diagram for Problem 4.23

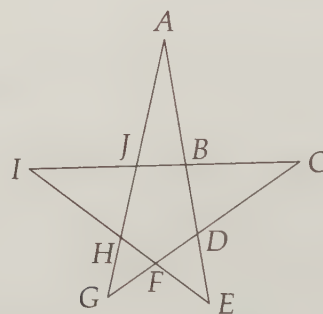


Figure 4.8: Diagram for Problem 4.24

4.24 Each of the outer triangles in the figure shown at right above has perimeter 23. The perimeter of $BDFHJ$ is 51. What is the perimeter of star $ABCDEFGHJI$? **Hints:** 510

4.25 In the figure at left below, $AB = 12$ cm and $BC = AD = 8$ cm. $\overline{BC} \perp \overline{AB}$ and $\overline{DA} \perp \overline{AB}$. How many square centimeters are in the area of the shaded region? (Source: MATHCOUNTS)



Figure 4.9: Diagram for Problem 4.25

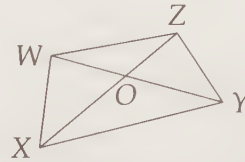


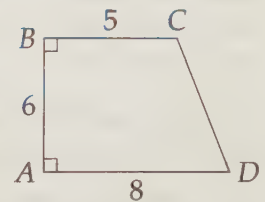
Figure 4.10: Diagram for Problem 4.26

4.26 In the figure at right above, $[WOZ] = 12$, $[ZOY] = 18$, and $[WXYZ] = 50$. Find $[WOX]$.

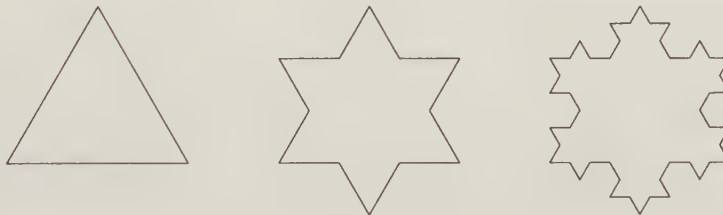
Challenge Problems

4.27 Find the area of $ABCD$ shown at right. **Hints:** 577

4.28 A gardener plans to build a fence to enclose a square garden plot. The perimeter of the plot is 96 feet, and he sets posts at the corners of the square. The posts along the sides are set 6 feet apart. How many posts will he use to fence the entire plot? (Source: MATHCOUNTS)



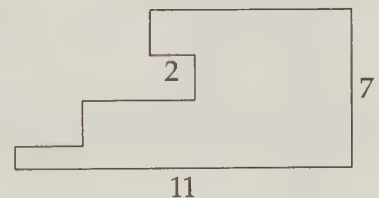
4.29 We begin with an equilateral triangle. We divide each side into three segments of equal length, and add an equilateral triangle to each side using the middle third as a base. We then repeat this, to get a third figure.



Given that the perimeter of the first figure is 12, what is the perimeter of the second figure? What is the perimeter of the third figure? **Hints:** 273, 141

4.30★ Suppose we continue the process described in Problem 4.29 forever. What is the perimeter of the resulting figure?

4.31 All sides of the building shown at right meet at right angles. If three of the sides measure 2 meters, 7 meters, and 11 meters as shown, then what is the perimeter of the building in meters? (Source: Mandelbrot)



4.32 In $\triangle ABC$ at left below, $CX = 2BX$ and $AY = 3BY$. Find $[BXY]$ given that $[ABC] = 144$. **Hints:** 285, 89

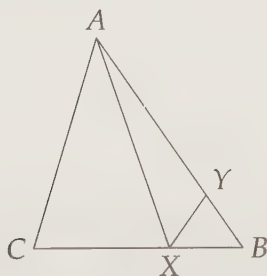


Figure 4.11: Diagram for Problem 4.32

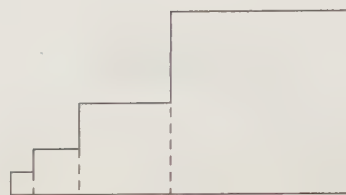


Figure 4.12: Diagram for Problem 4.33

4.33 The diagram at right above is formed by placing 4 squares together along a single line as shown. Each square has a side length that is $1/2$ the side length of the next larger square. The outer perimeter of the figure is 115. What is the area of the whole figure? (Source: ARML) **Hints:** 329

4.34 Sue and Barry are trying to find the area of $\triangle ABC$. Sue mistakenly uses AB and the height from A (instead of the height from C), and Barry mistakenly uses BC and the height from C (instead of the height from A).

- (a) Sue finds an area of 12 and Barry finds an area of 27. What is the area of $\triangle ABC$? **Hints:** 369, 38
 (b)★ Suppose instead that Sue finds an area of 120 and Barry finds an area of 150. Now what is the area of $\triangle ABC$? **Hints:** 301

4.35 Show that $[WPX] = [WPZ]$ in the diagram at left below. **Hints:** 426, 108

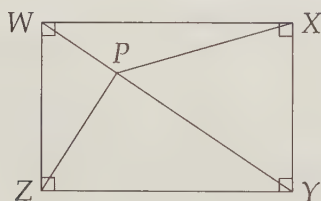


Figure 4.13: Diagram for Problem 4.35

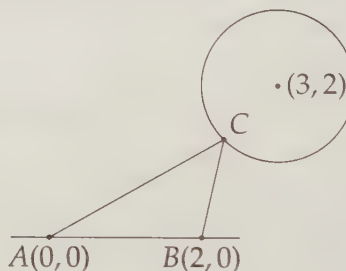


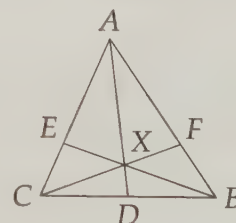
Figure 4.14: Diagram for Problem 4.36

4.36 In the diagram at right above, triangle ABC has vertex A at the origin, vertex B at the point $(2,0)$, and vertex C on the circle with center $(3,2)$ and radius 1. What is the maximum possible area for such a triangle? (Source: Mandelbrot) **Hints:** 75, 553

4.37★ $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$ meet at X as shown at right. Prove that

$$\frac{[AXC]}{[BXC]} = \frac{AF}{FB}.$$

Hints: 231, 98



4.38★ In the diagram at left below, given $[PQRS] = 5[PQA]$ and $[PQRS] = 4[PBS]$, find $[ABP]/[PQRS]$.
Hints: 583, 470, 361

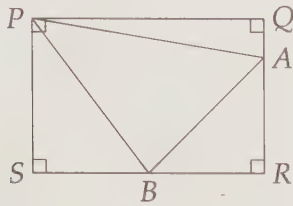


Figure 4.15: Diagram for Problem 4.38

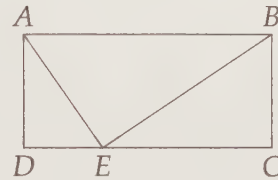
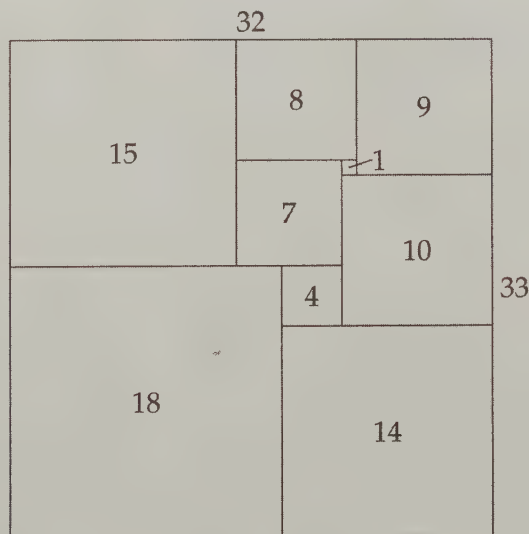


Figure 4.16: Diagram for Problem 4.39

4.39 In rectangle $ABCD$ at right above, point E is on side $\overline{CD}$ such that $[CBE] - [ADE] = [AEB] - [CBE]$. What is the ratio of the area of the largest region to the area of the smallest region? (Source: ARML)
Hints: 324, 166

4.40★ For Joe's birthday, Will has bought a 7×7 inch square birthday cake with a flat top. It is a chocolate cake with banana frosting on the top and on the sides. It turns out that seven people will be present when the cake is cut, and each person will become quite envious if another person receives more cake or frosting. Find a way to divide the cake among seven people, so that each receives an equal amount of cake and frosting. **Hints:** 149, 67, 124

Extra! Perhaps surprisingly, it is possible to dissect some rectangles into squares of different sizes. The figure below shows a dissection of a 32×33 rectangle into squares, whose side lengths are indicated.

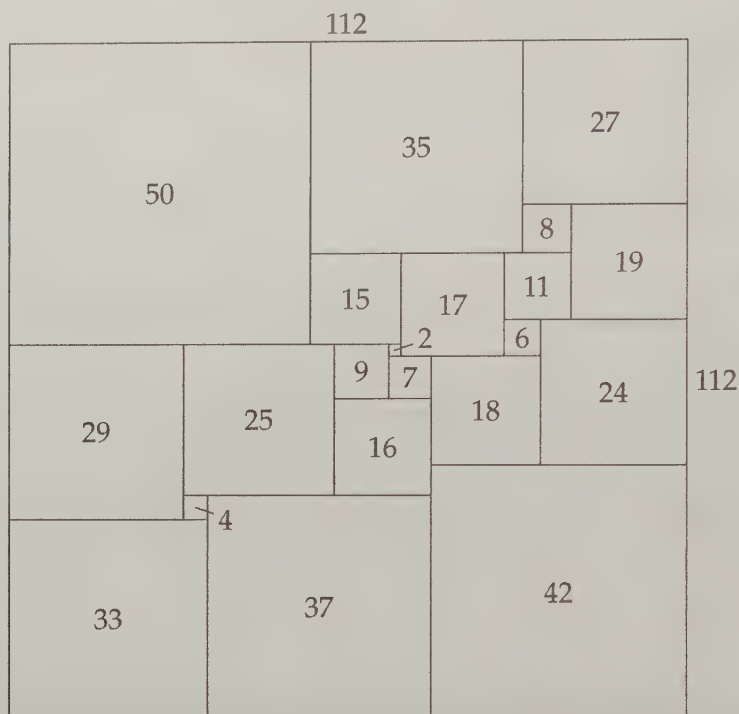


Is it possible to dissect a square into squares of different sizes?

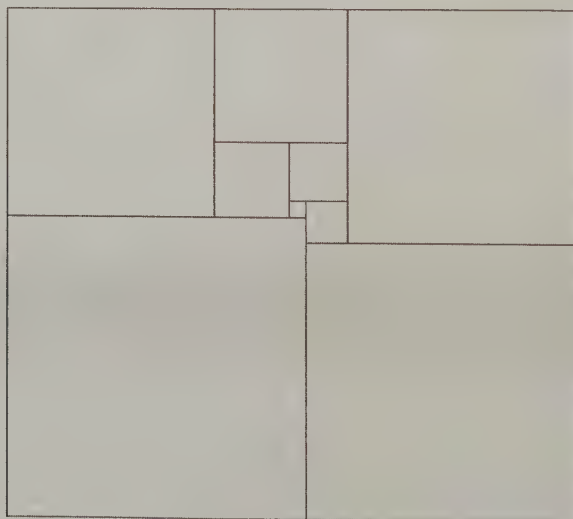
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Extra! . . . continued from the previous page

◆◆◆◆◆ The answer is yes! The figure below shows such a dissection of a 112×112 square. (The numbers indicate side lengths of the squares.)



A problem from the 2000 American Invitational Mathematics Examination asks the following: "The diagram shows a rectangle that has been dissected into nine non-overlapping squares. Given that the width and the height of the rectangle are positive integers with greatest common divisor 1, find the perimeter of the rectangle."



Try to solve it!